

SIMATS ENGINEERING

ASSESSMENT TOOL 1: CLASS TEST

Course Name: Cloud Computing and Big Data Analytics

Course Code: CSA1516

Course Outcome Covered: CO4 – Analyze big data characteristics and challenges across domains including security and application aspects (BL4)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Articulation (Level 4)

Question Count: 10

Total Marks: 30

Student Details

Name	Sanin Febin
Register Number	192511169
Assessment	Class Test – Big Data Analytics

Code of Conduct

I R.Jaideep (Reg No: 192525441) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R.Jaideep

CLASS TEST – ANSWERS

1. Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.

Big Data refers to extremely large, rapidly generated, diverse, and complex datasets that cannot be efficiently captured, stored, processed, or analyzed using conventional data-processing systems. Big Data is commonly characterized by the 5Vs: Volume, Velocity, Variety, Veracity, and Value.

Traditional systems are often insufficient because they are designed for limited data volumes and relatively stable, structured data. They may not scale efficiently when data grows to terabytes or petabytes. They can also struggle with high-speed data streams, unstructured data such as videos and social-media posts, and the need for distributed processing. Big data platforms therefore use distributed storage and parallel processing technologies such as Hadoop and Spark to divide large workloads across multiple machines.

2. Explain the 5Vs of Big Data with one practical example for any two Vs.

The 5Vs of Big Data are:

- Volume – Refers to the huge quantity of data generated and stored. Example: An e-commerce company may store millions of customer transactions, product views, and click records.
- Velocity – Refers to the speed at which data is generated, transmitted, and processed. Example: Stock-market systems receive and process thousands of price updates every second.
- Variety – Refers to different forms of data, including structured, semi-structured, and unstructured data. Example: A hospital may manage database records, XML/JSON reports, medical images, and doctor notes.
- Veracity – Refers to the accuracy, reliability, and quality of data. Incorrect or duplicated customer records can reduce the reliability of analytics.
- Value – Refers to the useful insights and benefits obtained from data. For example, analyzing customer purchases can help a retailer recommend suitable products and improve sales.

3. Differentiate structured, semi-structured, and unstructured data with examples.

Structured data has a fixed schema and is usually organized in rows and columns. It can be easily stored and queried using relational databases. Examples include student records, bank transactions, and employee tables.

Semi-structured data does not follow a strict tabular schema but contains organizational elements such as tags, keys, or metadata. Examples include JSON, XML, and email data.

Unstructured data has no predefined data model or fixed tabular structure. Examples include photographs, videos, audio files, social-media posts, scanned documents, and free-form text.

Thus, structured data is easiest to process using traditional databases, while semi-structured and unstructured data commonly require flexible storage and specialized big data processing techniques.

4. Explain the role of distributed file systems in big data processing.

A distributed file system stores large files by dividing them into blocks and distributing those blocks across multiple machines or nodes. This allows storage capacity and processing workloads to be distributed across a cluster.

In big data environments, distributed file systems such as the Hadoop Distributed File System (HDFS) provide scalability, fault tolerance, and high-throughput access. Data is replicated across multiple nodes so that the system can continue operating even if one node fails. Distributed storage also supports parallel processing because different parts of a dataset can be processed simultaneously. Therefore, distributed file systems form an important foundation for storing and analyzing very large datasets.

5. List any three major security concerns in big data platforms and briefly explain each.

1. Data Privacy and Unauthorized Access – Big data platforms may contain sensitive personal, financial, or business information. Weak authentication or access control can expose this information.
2. Data Breaches and Leakage – Large centralized datasets can become attractive targets for attackers. A breach can result in the loss or disclosure of huge quantities of information.
3. Data Integrity and Tampering – Attackers or unauthorized users may alter data. Incorrect data can produce unreliable analytical results and lead to poor decisions.

Other concerns include insecure APIs, insider threats, inadequate encryption, compliance issues, and poor monitoring. Security can be improved using encryption, authentication, authorization, auditing, access controls, and secure data-governance practices.

6. What are data appliances and integration tools? Explain their role in big data workflows.

Data appliances are specialized hardware or integrated systems designed to store, process, manage, or analyze large volumes of data efficiently. They may combine storage, computing, networking, and database technologies in a preconfigured platform.

Data integration tools are software solutions that collect data from multiple sources, transform it into usable formats, and move it into storage or analytics platforms. They support processes such as ETL (Extract, Transform, Load) and data pipelines.

In a big data workflow, these technologies help organizations connect databases, applications, cloud services, logs, sensors, and other sources. They improve data movement, transformation, consistency, and availability for analytics. Examples of integration technologies include Apache Kafka, Apache NiFi, and ETL platforms.

7. Discuss two major challenges organizations face while implementing big data analytics.

1. Data Quality and Integration – Organizations collect data from databases, applications, sensors, websites, and external sources. The data may have missing values, duplicates, inconsistent formats, or errors. Integrating and cleaning such data requires significant effort.

2. Security, Privacy, and Governance – Big data may contain sensitive customer and organizational information. Organizations must control who can access data, protect it from breaches, comply with regulations, and maintain proper data governance.

Other challenges include infrastructure cost, shortage of skilled professionals, scalability, complex architecture, and difficulty in converting analytical results into business value. Successful implementation therefore requires suitable technology, skilled personnel, security controls, and clear business objectives.

8. Write short notes on any one domain application of big data analytics.

Healthcare is an important application domain for big data analytics. Hospitals and healthcare organizations generate data from electronic health records, laboratory reports, medical images, wearable devices, prescriptions, and patient-monitoring systems.

Big data analytics can help identify disease patterns, predict patient risks, support personalized treatment, improve hospital resource planning, and monitor public-health trends. For example, analyzing historical patient data can help identify factors associated with readmission or disease progression.

However, healthcare analytics must carefully protect patient privacy and maintain data security. Proper access control, encryption, anonymization, and governance are essential when handling sensitive medical information.

9. How do business drivers influence the adoption of big data solutions?

Business drivers are the organizational needs and opportunities that motivate companies to adopt big data solutions. A major driver is the need for better decision making. Analytics can convert large datasets into useful insights for managers.

Other drivers include improving customer experience, reducing operational costs, identifying new revenue opportunities, detecting fraud, predicting demand, and gaining competitive advantage. For example, an online retailer can analyze customer browsing and purchasing behavior to provide personalized recommendations.

Therefore, organizations are more likely to invest in big data when the expected business value is clear. Technology adoption should be aligned with measurable objectives such as increased revenue, reduced cost, improved efficiency, or better customer satisfaction.

10. Explain the importance of big data analytics in decision making.

Big data analytics supports decision making by converting large and complex datasets into meaningful information and actionable insights. Instead of relying only on assumptions or historical intuition,

organizations can use data-driven evidence to understand current conditions and predict future outcomes.

Analytics helps identify trends, customer behavior, operational problems, risks, and opportunities. Descriptive analytics explains what happened, diagnostic analytics helps identify why it happened, predictive analytics estimates what may happen, and prescriptive analytics can recommend suitable actions.

For example, a retailer can analyze sales, customer behavior, inventory, and seasonal data to forecast demand and optimize stock levels. Thus, big data analytics improves the speed, accuracy, and quality of strategic and operational decisions.

CLASS TEST RUBRIC

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

Prepared for: Sanin Febin | Reg No: 192511169